

The Fabius Function and Rvachev's up-Function

Analysis, exact dyadic arithmetic, and the corrected full endpoint asymptotic expansion

A human-readable exposition based on the Lean formalization
in Vladimir Reshetnikov's `ProveIt/Analysis/FabiusFunction`

24 August 2026

Abstract

The Fabius function is the unique smooth distribution function on the unit interval satisfying the dyadic differential equation $F'(x) = 2F(2x)$ on the left half of the interval and the reflection law $F(1 - x) = 1 - F(x)$. Its even, compactly supported twin is Rvachev's up-function, the distinguished C^∞ bump on $[-1, 1]$ satisfying

$$\text{up}'(x) = 2(\text{up}(2x + 1) - \text{up}(2x - 1)), \quad \text{up}(0) = 1.$$

A third, signed global extension satisfies the simpler scaling equation $\Theta'(x) = 2\Theta(2x)$. Keeping these three objects separate resolves a number of notational ambiguities in the literature.

This article develops the principal results formalized in Lean in the `ProveIt` repository. We give constructions by infinite convolution, random series, Fourier products, and step functions; prove smoothness, positivity, support, partition-of-unity, Poisson, moment, derivative, and non-analyticity results; derive exact rational formulas at every dyadic point; explain the fast bit-recursive evaluator and the finite q -binomial–Thue–Morse formula; and treat the associated integer, denominator, and parity phenomena.

Particular attention is paid to the endpoint $x \downarrow 0$. The logarithm of $F(x)$ has a quadratic logarithmic scale, but its sharp main term contains a genuine one-periodic fluctuation. We derive its exact mean and its absolutely convergent Fourier series with Gamma–zeta coefficients, introduce the exact lower-Lambert phase $\lambda 2^{-\lambda} = x$, and state the full all-orders expansion in inverse powers of λ . The first correction is given explicitly. This also explains why a commonly repeated purely elementary asymptotic, with the periodic term omitted, cannot have a vanishing error.

Contents

1	Introduction and conventions	4
1.1	A guide to the main results	4
2	The three related functions	5
2.1	The bounded Fabius function	5
2.2	Rvachev's compactly supported twin	5
2.3	The signed global extension	6
3	Construction by probability, convolution, and Fourier transform	6
3.1	An infinite random series	6
3.2	Infinite convolution	7
3.3	The distribution-function interpretation	8

4	Step approximants and elementary structure	8
4.1	A sequence of combinatorial step functions	8
4.2	Partition of unity and Poisson summation	9
4.3	Cosine and Weierstrass products	10
5	The global extension and the derivative tower	11
6	Moments, generating functions, and inverse powers of two	12
6.1	Even moments	12
6.2	Half-moments and a generating function	13
7	A closed formula for every dyadic value	14
7.1	Examples	15
7.2	Refinement invariance	16
8	A fast bit-recursive exact evaluator	16
8.1	Taylor data at inverse powers of two	16
8.2	Precomputation at inverse powers	16
8.3	The binary algorithm	17
8.4	Bounded, global, and rational interfaces	17
9	The finite q-binomial–Thue–Morse formula	18
10	Arithmetic sequences, denominators, and parity	19
10.1	Integer normalizations of the moments	19
10.2	Reshetnikov's integers	20
10.3	Denominator bounds for all dyadic values	20
10.4	Lucas parity and exact 2-adic valuations	21
10.5	Bernoulli recurrences	22
11	The Fourier–Legendre expansion	22
	Endpoint asymptotics	23
12	The negative Laplace product and dyadic renormalization	23
12.1	The exact Laplace identity	23
12.2	An exact periodic decomposition	24
13	The Gamma–zeta periodic fluctuation	25
14	The exact lower-Lambert phase	26
15	The sharp endpoint main term	26
15.1	The leading quadratic law	28
16	Corrected explicit forms and the failure of the nonperiodic formula	28
16.1	The compact Lambert– W form	28
16.2	An elementary logarithmic expansion	28
16.3	Dyadic logarithmic scale	29
16.4	Why the periodic term cannot be omitted	29

17 The first saddle correction	29
18 The full all-orders expansion	30
18.1 All-order expansion of the phase	31
18.2 A numerical check at exact inverse powers	32
19 Consequences and conceptual summary	32
A Infinite binary-reduction series and discrete limits	33
A.1 Binary prefixes, tails, and the analytic telescope	33
A.2 The parity-power form and the missing endpoint term	34
A.3 The fully explicit global q -binomial series	35
A.4 A finite discrete approximation converging to Θ	35
B Concordance with the formal development and source corrections	36
B.1 Corrections incorporated in the statements	37

1 Introduction and conventions

The Fabius function is remarkable for combining features that normally live in rather different parts of analysis. It is C^∞ but nowhere real-analytic on its active interval; it is governed by a self-similar functional-differential equation; its values at dyadic rationals are exactly rational; its Fourier transform is an elementary infinite product; its moments conceal striking integer sequences; and its endpoint behavior is controlled by a Lambert– W saddle together with a tiny but indispensable periodic oscillation.

The name “Fabius function” is used in two incompatible ways. Some authors mean a distribution function F on $[0, 1]$, while others use the same letter for an unbounded signed solution of $f'(x) = 2f(2x)$. The compactly supported Rvachev function is then introduced by a shift and a reflection, sometimes without a change of notation. We shall instead use three symbols throughout:

symbol	object	normalization/domain convention
F	bounded Fabius function	$F = 0$ on $(-\infty, 0]$, $F = 1$ on $[1, \infty)$
up	Rvachev bump	even, supported on $[-1, 1]$, $\text{up}(0) = 1$
Θ	signed global extension	$\Theta = F$ on $[0, 1]$, $\Theta' = 2\Theta(2\cdot)$

This convention follows the formal development and prevents identities valid for Θ from being mistakenly asserted for the bounded function F .

We use the Fourier transform

$$\widehat{g}(z) = \int_{\mathbb{R}} g(t) e^{-2\pi i z t} dt. \quad (1.1)$$

The binary weight $\text{wt}(n)$ is the number of 1's in the binary expansion of n , and

$$\epsilon_n = (-1)^{\text{wt}(n)} \quad (1.2)$$

is the Thue–Morse sign. We write $L = \log 2$. The Stieltjes constants are normalized by

$$\zeta(1+z) = \frac{1}{z} + \sum_{m \geq 0} \frac{(-1)^m \gamma_m}{m!} z^m, \quad \gamma_0 = \gamma.$$

1.1 A guide to the main results

The structural equivalence between F and up is established in [Section 2](#). The random-series and Fourier constructions are in [Section 3](#), and the basic analytic properties are in [Section 4](#). The global extension and its derivative tower appear in [Section 5](#). Moments and generating functions are handled in [Section 6](#).

The exact arithmetic begins in [Section 7](#). The central formula is [Theorem 7.2](#), which evaluates every dyadic value by a finite Thue–Morse sum and finitely many even moments. A second evaluator, based on terminating Taylor recursion and the binary digits of the numerator, is given in [Section 8](#). The much less obvious q -binomial formula is proved in [Section 9](#). Integrality, denominator bounds, and parity are collected in [Section 10](#).

The endpoint analysis begins in [Section 12](#). The exact dyadic renormalization of the negative Laplace product produces a periodic function Ψ with explicit Gamma–zeta Fourier coefficients. The lower-Lambert phase is introduced in [Section 14](#); the sharp main theorem is [Theorem 15.1](#); and the all-orders expansion is [Theorem 18.1](#). The corrected elementary and Lambert– W forms are displayed in [Section 16](#).

2 The three related functions

2.1 The bounded Fabius function

Definition 2.1. A *bounded Fabius function* is a function $F : \mathbb{R} \rightarrow [0, 1]$ such that

- (i) $F(x) = 0$ for $x \leq 0$ and $F(x) = 1$ for $x \geq 1$;
- (ii) $F \in C^\infty(\mathbb{R})$;
- (iii) $F(1 - x) = 1 - F(x)$ for $0 \leq x \leq 1$;
- (iv) $F'(x) = 2F(2x)$ for $0 \leq x \leq \frac{1}{2}$.

The endpoint constancy and smoothness force every derivative of positive order to vanish at 0 and 1. The reflection law gives $F(\frac{1}{2}) = \frac{1}{2}$. Differentiating the reflection identity and using the left-half equation shows that on the right half

$$F'(x) = 2F(2 - 2x), \quad \frac{1}{2} \leq x \leq 1. \quad (2.1)$$

Thus F is increasing, with a symmetric density about $1/2$.

2.2 Rvachev's compactly supported twin

Definition 2.2. Given F , define

$$\text{up}(x) = \begin{cases} F(x + 1), & x \leq 0, \\ F(1 - x), & x \geq 0. \end{cases} \quad (2.2)$$

The two pieces agree to all orders at 0. Indeed they have the common value 1, and their positive-order derivatives are related by the symmetry of the density. The function is even, supported on $[-1, 1]$, positive in the open interval, and normalized by $\text{up}(0) = 1$.

Theorem 2.3 (Rvachev equation). *The function up satisfies*

$$\text{up}'(x) = 2(\text{up}(2x + 1) - \text{up}(2x - 1)) \quad (x \in \mathbb{R}). \quad (2.3)$$

Conversely, there is a unique even C^∞ function supported on $[-1, 1]$ with integral 1 that satisfies (2.3); it has $\text{up}(0) = 1$ and gives the bounded Fabius function by

$$F(x) = \text{up}(x - 1), \quad 0 \leq x \leq 1, \quad (2.4)$$

with constant continuation outside the unit interval.

Proof. For $-1 < x < 0$, (2.2) and the two half-interval formulas for F' give

$$\text{up}'(x) = F'(x + 1) = 2\text{up}(2x + 1),$$

because $2x - 1 < -1$ and hence $\text{up}(2x - 1) = 0$. The case $0 < x < 1$ follows by evenness, and outside $[-1, 1]$ both sides vanish. Smoothness extends the identity through the finitely many boundary points.

Existence and uniqueness are proved below by Fourier transform. Once $\int \text{up} = 1$ is known, integration of (2.3) from -1 to 0 gives

$$\text{up}(0) = 2 \int_{-1}^0 \text{up}(2t + 1) dt = \int_{-1}^1 \text{up}(u) du = 1.$$

On $[-1, 0]$ the same integration gives

$$\text{up}(x) = 2 \int_{-1}^x \text{up}(2t + 1) dt. \quad (2.5)$$

It follows that $F(x) = \text{up}(x - 1)$ obeys $F'(x) = 2 \text{up}(2x - 1) = 2F(2x)$ on the left half. The partition identity proved in [Theorem 4.4](#) gives the reflection law. Hence the two descriptions are equivalent. $\square$

2.3 The signed global extension

The simplest differential equation belongs not to the bounded F but to a signed global function. Define

$$\Theta(x) = \sum_{m=0}^{\infty} \epsilon_m \text{up}(x - 2m - 1). \quad (2.6)$$

Because up has compact support, the sum is locally finite. On the first unit interval only the term $m = 0$ is present, and therefore

$$\Theta(x) = \text{up}(x - 1) = F(x), \quad 0 \leq x \leq 1. \quad (2.7)$$

The function Θ is zero on the negative half-line, is alternately reflected and signed on consecutive blocks of length 2, and is not bounded by the right-hand constant convention that defines F .

Warning 2.4. The equation $f'(x) = 2f(2x)$ is valid globally for Θ , not for the bounded function F . On $[0, \frac{1}{2}]$ they agree and hence the familiar Fabius equation holds there, but for $x \geq 1$ the bounded function is constant while Θ continues to oscillate in signed compact arcs.

3 Construction by probability, convolution, and Fourier transform

3.1 An infinite random series

Let $V_1, V_2, \dots$ be independent random variables uniformly distributed on $[-1, 1]$, and set

$$Y = \sum_{j=1}^{\infty} \frac{V_j}{2^j}. \quad (3.1)$$

The series converges absolutely almost surely and $|Y| \leq 1$. Its characteristic function, in the convention (1.1), is

$$\mathbb{E} e^{-2\pi i z Y} = \prod_{j=1}^{\infty} \frac{\sin(2\pi z / 2^j)}{2\pi z / 2^j} = \prod_{h=0}^{\infty} \frac{\sin(\pi z / 2^h)}{\pi z / 2^h}. \quad (3.2)$$

The product will be denoted by $\Phi(z)$.

Lemma 3.1 (Rapid decay of the product). *For every integer $N \geq 0$ there is C_N such that*

$$|\Phi(\xi)| \leq C_N (1 + |\xi|)^{-N}, \quad \xi \in \mathbb{R}.$$

The same is true after multiplying by any fixed power of ξ .

Proof. The elementary estimate $|\sin u/u| \leq \min(1, |u|^{-1})$ implies that, once $|\xi|$ is large, any chosen N initial factors for which $|\pi\xi/2^h| \geq 1$ contribute at most a constant times $|\xi|^{-N}$; the remaining factors have modulus at most 1. The range of small ξ is absorbed into the constant. Taking more factors proves the second statement. $\square$

Fourier inversion therefore gives a Schwartz-class density

$$\text{up}(x) = \int_{\mathbb{R}} \Phi(\xi) e^{2\pi i x \xi} d\xi. \quad (3.3)$$

The law of Y is supported on $[-1, 1]$, so the density vanishes outside this interval. Symmetry of Y makes up even, and $\int \text{up} = \Phi(0) = 1$. Since Φ is rapidly decreasing, differentiation under the integral shows that $\text{up} \in C^\infty$.

Theorem 3.2 (Existence and uniqueness). *The density in (3.3) is the unique normalized C^∞ solution of the Rvachev equation. Consequently the bounded Fabius function exists and is unique.*

Proof. The product satisfies

$$\Phi(z) = \frac{\sin \pi z}{\pi z} \Phi(z/2). \quad (3.4)$$

Taking inverse Fourier transforms yields (2.3); alternatively, one can obtain the equation directly by conditioning on V_1 and differentiating the resulting convolution integral.

Conversely, if g is an integrable normalized solution of (2.3), Fourier transformation gives

$$\widehat{g}(z) = \frac{\sin \pi z}{\pi z} \widehat{g}(z/2).$$

Iterating n times gives

$$\widehat{g}(z) = \prod_{h=0}^{n-1} \frac{\sin(\pi z/2^h)}{\pi z/2^h} \widehat{g}(z/2^n).$$

Since $\widehat{g}(0) = 1$ and $\widehat{g}$ is continuous at the origin, letting $n \rightarrow \infty$ gives $\widehat{g} = \Phi$. Injectivity of the Fourier transform then gives $g = \text{up}$. $\square$

3.2 Infinite convolution

Let $b_a(x) = (2a)^{-1} \mathbf{1}_{[-a, a]}(x)$ be the centered uniform density. Then

$$\text{up} = b_{1/2} * b_{1/4} * b_{1/8} * \cdots \quad (3.5)$$

in the sense of weak convergence of probability measures, and in fact in C^m after enough factors for each fixed m . Formula (3.5) makes nonnegativity immediate. It also gives an efficient conceptual proof of compact support: the Minkowski sum of the supports has radius $\sum_{j \geq 1} 2^{-j} = 1$.

Strict positivity in $(-1, 1)$ can be seen probabilistically. Given $|x| < 1$, choose finitely many coordinates of the random series in small open intervals whose weighted sum lies near x ; the remaining tail has arbitrarily small support. The resulting cylinder event has positive probability, so every sufficiently small interval about x has positive mass. Continuity and the differential identity then rule out an interior zero. Equivalently, (2.5), evenness, and an elementary propagation argument show that positivity at one point propagates to the whole open support.

3.3 The distribution-function interpretation

Let $U_1, U_2, \dots$ be independent uniform random variables on $[0, 1]$ and put

$$X = \sum_{j=1}^{\infty} \frac{U_j}{2^j}. \quad (3.6)$$

Writing $V_j = 2U_j - 1$ gives $Y = 2X - 1$. The following identity is one of the most useful ways to understand the Fabius function.

Theorem 3.3 (Probability representation). *For $0 \leq x \leq 1$,*

$$F(x) = \mathbb{P}\{X \leq x\} = \text{up}(x - 1). \quad (3.7)$$

The density of X is

$$F'(x) = 2 \text{up}(2x - 1). \quad (3.8)$$

Proof. Since $Y = 2X - 1$, the density transformation gives the density $2 \text{up}(2x - 1)$ for X . Hence its distribution function is

$$\int_0^x 2 \text{up}(2t - 1) dt = \int_{-1}^{2x-1} \text{up}(u) du.$$

On the other hand, integrating the Rvachev equation on $[-1, x - 1]$ gives

$$\text{up}(x - 1) = 2 \int_{-1}^{x-1} \text{up}(2t + 1) dt = \int_{-1}^{2x-1} \text{up}(u) du.$$

Thus the two expressions agree. $\square$

The support and symmetry of X are now transparent. Since replacing every U_j by $1 - U_j$ replaces X by $1 - X$, the distribution is symmetric about $1/2$, which immediately gives $F(1 - x) = 1 - F(x)$.

4 Step approximants and elementary structure

4.1 A sequence of combinatorial step functions

Define polynomials p_n by

$$p_0(X) = 1, \quad p_n(X) = p_{n-1}(X^2)(1 + X)^n. \quad (4.1)$$

A simple induction gives the two useful product forms

$$p_n(X) = \prod_{j=1}^n \frac{1 - X^{2^j}}{1 - X} = \prod_{j=1}^n (1 + X + \dots + X^{2^j-1}). \quad (4.2)$$

Write

$$p_n(X) = \sum_{m=0}^{g_n} a_{n,m} X^m.$$

Then $g_0 = 0$, $g_n = 2g_{n-1} + n$, and

$$\frac{g_n}{2^n} = \sum_{j=1}^n \frac{j}{2^j} \longrightarrow 2. \quad (4.3)$$

The coefficient $a_{n,m}$ counts representations

$$m = s_1 + \cdots + s_n, \quad 0 \leq s_j \leq 2^j - 1.$$

For an interval $I = [a, b]$, let $\mathbf{1}_I^*$ be 1 in its interior, $1/2$ at either endpoint, and 0 elsewhere. The half-weight convention is important: adjacent intervals share endpoints, and using ordinary closed indicators would double-count those points. Put

$$\text{up}_n(x) = 2^{n - \binom{n+1}{2}} \sum_{m=0}^{g_n} a_{n,m} \mathbf{1}_{I_{n,m}}^*(x), \quad (4.4)$$

where

$$I_{n,m} = \left[\frac{2m-1-g_n}{2^{n+1}}, \frac{2m+1-g_n}{2^{n+1}} \right]. \quad (4.5)$$

Theorem 4.1 (Step-function approximation). *The functions up_n are even, unimodal, satisfy $\text{up}_n(0) = 1$, and converge pointwise to up on $\mathbb{R}$. Equivalently, the associated finite probability measures converge weakly to the probability measure with density up .*

Proof. Normalize p_n by $2^{-\binom{n+1}{2}}$. Its coefficients sum to 1, so replacing X^m by a point mass at

$$\frac{2m - g_n}{2^{n+1}}$$

produces a probability measure μ_n . Formula (4.2) identifies this measure with the law of a finite sum of independent symmetric lattice variables. Its characteristic function is the corresponding finite partial product in (3.2); hence μ_n converges weakly to the law with density up .

The step function (4.4) is obtained by spreading every atom uniformly over a cell of width 2^{-n} . Integrals of up_n against indicators of intervals with dyadic endpoints therefore agree with the corresponding masses up to a boundary term that tends to zero. Weak convergence gives convergence of these integrals to those of up . Symmetry and the log-concave/unimodal coefficient recurrence inherited from (4.1) show that up_n is nondecreasing on $(-1, 0)$ and nonincreasing on $(0, 1)$. Since the limit is continuous, convergence of the interval integrals plus monotonicity gives pointwise convergence. The normalization at zero follows directly from the central coefficient recurrence. $\square$

Remark 4.2. The step approximation is not merely a picture-making device. It gives a finite combinatorial model for up , makes positivity visible before any Fourier analysis, and is particularly convenient for certified numerical bounds because all breakpoints and heights are dyadic rationals times integers.

4.2 Partition of unity and Poisson summation

The rapid decay of $\widehat{\text{up}}$ permits Poisson summation without any regularization.

Theorem 4.3 (Poisson formula). *For $u > 0$ and $t \in \mathbb{R}$,*

$$\sum_{k \in \mathbb{Z}} \text{up}(t + uk) = \frac{1}{u} \sum_{m \in \mathbb{Z}} \widehat{\text{up}}(m/u) e^{2\pi i m t / u}. \quad (4.6)$$

Both series converge absolutely after the usual interpretation of the locally finite left side.

Proof. Apply the classical Poisson formula to the Schwartz function up . Although up is compactly supported rather than rapidly decaying in the usual nontrivial way, it is still a Schwartz function because every derivative is compactly supported. Our Fourier convention gives exactly the phase in (4.6). $\square$

Proposition 4.4 (Dyadic and integral partitions of unity). *For every positive integer n ,*

$$\sum_{k \in \mathbb{Z}} \text{up}\left(t + \frac{k}{n}\right) = n. \quad (4.7)$$

In particular,

$$\sum_{k \in \mathbb{Z}} \text{up}(t + k) = 1, \quad \text{up}(t) + \text{up}(t - 1) = 1 \quad (0 \leq t \leq 1). \quad (4.8)$$

Proof. Take $u = 1/n$ in (4.6). For every nonzero integer m , $\widehat{\text{up}}(nm) = 0$, because the first sinc factor in (3.2) vanishes. Only the zero Fourier mode remains and has value $n\widehat{\text{up}}(0) = n$. The support restriction reduces the case $n = 1$ to the two-term identity on $[0, 1]$. $\square$

The last identity is another direct proof that $F(x) = \text{up}(x - 1)$ satisfies $F(1 - x) = 1 - F(x)$.

4.3 Cosine and Weierstrass products

The elementary identity

$$\frac{\sin z}{z} = \prod_{j=1}^{\infty} \cos(z/2^j)$$

and a rearrangement of absolutely convergent products give

$$\widehat{\text{up}}(z) = \prod_{m=1}^{\infty} \cos(\pi z/2^m)^m. \quad (4.9)$$

Combining this with Euler's product for the sine gives a second factorization:

$$\widehat{\text{up}}(z) = \prod_{r=1}^{\infty} \left(1 - \frac{z^2}{r^2}\right)^{1+\nu_2(r)}, \quad (4.10)$$

where $\nu_2(r)$ is the exponent of 2 in r . Indeed, a fixed integer r occurs once for every representation $r = 2^h m$, namely $0 \leq h \leq \nu_2(r)$.

Taking $u = 2$ in Poisson summation, using support and pairing conjugate modes, we obtain the uniformly convergent cosine expansion on $[-1, 1]$:

$$\text{up}(t) = \frac{1}{2} + \sum_{m=0}^{\infty} \widehat{\text{up}}\left(\frac{2m+1}{2}\right) \cos((2m+1)\pi t). \quad (4.11)$$

All nonzero even Fourier samples disappear because they are integer zeros of $\widehat{\text{up}}$. Formula (4.11) is useful for high precision evaluation away from the endpoints; exact dyadic evaluation is better handled arithmetically in Section 7.

5 The global extension and the derivative tower

We return to the locally finite series

$$\Theta(x) = \sum_{m \geq 0} \epsilon_m \text{up}(x - 2m - 1), \quad \epsilon_m = (-1)^{\text{wt}(m)}. \quad (5.1)$$

The binary identities

$$\epsilon_{2m} = \epsilon_m, \quad \epsilon_{2m+1} = -\epsilon_m \quad (5.2)$$

are exactly what is needed to turn the two translates in the Rvachev equation into one scaled copy of Θ .

Theorem 5.1 (Global scaling equation). *The function Θ is C^∞ and*

$$\Theta'(x) = 2\Theta(2x), \quad x \in \mathbb{R}. \quad (5.3)$$

For every $k \geq 0$,

$$\Theta^{(k)}(x) = 2^{\binom{k+1}{2}} \Theta(2^k x). \quad (5.4)$$

Moreover, for $-1 \leq t \leq 1$,

$$\text{up}^{(k)}(t) = 2^{\binom{k+1}{2}} \Theta(2^k t + 2^k). \quad (5.5)$$

Proof. Local finiteness permits termwise differentiation. Using (2.3),

$$\begin{aligned} \Theta'(x) &= 2 \sum_{m \geq 0} \epsilon_m [\text{up}(2x - 4m - 1) - \text{up}(2x - 4m - 3)] \\ &= 2 \sum_{m \geq 0} \epsilon_{2m} \text{up}(2x - 2(2m) - 1) + 2 \sum_{m \geq 0} \epsilon_{2m+1} \text{up}(2x - 2(2m + 1) - 1) \\ &= 2\Theta(2x). \end{aligned}$$

Induction gives (5.4), since differentiating the k th identity contributes an additional factor 2^{k+1} . Finally, $\text{up}(t) = \Theta(t + 1)$ on $[-1, 1]$; differentiating and applying (5.4) at $t + 1$ proves (5.5). $\square$

At integers, the support of up makes the global extension especially simple:

$$\Theta(2m) = 0, \quad \Theta(2m + 1) = \epsilon_m \quad (m \geq 0). \quad (5.6)$$

These values explain both the terminating Taylor expansions at dyadic points and the exact-arithmetic formulas later in the article.

Corollary 5.2 (Terminating Taylor data at dyadic points). *Let $t = q/2^n \in (-1, 1)$ be in lowest dyadic terms, so q is odd. Then*

$$\text{up}^{(k)}(t) = 0 \quad (k > n), \quad \text{up}^{(n)}(t) \neq 0. \quad (5.7)$$

Consequently the formal Taylor series of up at t is the polynomial

$$T_t(h) = \sum_{k=0}^n \frac{\text{up}^{(k)}(t)}{k!} h^k. \quad (5.8)$$

All Taylor coefficients after degree n vanish.

Proof. For $k \geq n$, the argument of Θ in (5.5) is

$$2^k t + 2^k = 2^{k-n} q + 2^k.$$

At $k = n$ this is the odd integer $q + 2^n$, and (5.6) gives a nonzero value. For $k > n$ it is even, so the value is zero. $\square$

Theorem 5.3 (Nowhere analyticity). *The function up is not real-analytic at any point of $[-1, 1]$. Likewise, F is not real-analytic at any point of $[0, 1]$.*

Proof. Suppose up were analytic on a neighborhood of some point. That neighborhood contains a reduced dyadic point $q/2^n$. By Theorem 5.2, analyticity there would force up to agree on a smaller interval with a polynomial of degree at most n . The same interval contains reduced dyadic points of arbitrarily large denominator exponent N . At such a point the N th derivative is nonzero by (5.7), whereas every derivative of order greater than n of the polynomial vanishes. This contradiction proves the claim. The assertion for F follows by translation on each half of the unit interval. $\square$

Remark 5.4 (Flat but not analytic). At the boundary, $F^{(k)}(0) = 0$ for every $k \geq 0$, yet $F(x) > 0$ for every $x > 0$. Thus the Taylor series at zero is identically zero. The endpoint asymptotic in Section 12 quantifies this flatness much more precisely: it is essentially the exponential of a negative quadratic in $\log(1/x)$, modified by logarithmic and periodic terms.

6 Moments, generating functions, and inverse powers of two

6.1 Even moments

Define

$$c_n = \int_{-1}^1 t^{2n} \text{up}(t) dt, \quad n \geq 0. \quad (6.1)$$

Then $c_0 = 1$, and the Taylor series of the Fourier transform is

$$\widehat{\text{up}}(z) = \sum_{n=0}^{\infty} (-1)^n c_n \frac{(2\pi z)^{2n}}{(2n)!}. \quad (6.2)$$

Proposition 6.1 (Moment recurrence). *For $n \geq 1$,*

$$(2n+1)(2^{2n}-1)c_n = \sum_{k=0}^{n-1} \binom{2n+1}{2k} c_k. \quad (6.3)$$

The first values are

$$1, \quad \frac{1}{9}, \quad \frac{19}{675}, \quad \frac{583}{59535}, \quad \frac{132809}{32531625}, \quad \frac{46840699}{24405225075}, \dots \quad (6.4)$$

Proof. Insert (6.2) into the refinement equation (3.4), expand $\sin(\pi z)/(\pi z)$, and compare the coefficient of z^{2n} . Equivalently, coefficient comparison first gives

$$(2n+1)2^{2n}c_n = \sum_{k=0}^n \binom{2n+1}{2k} c_k.$$

The final summand is $(2n+1)c_n$; moving it to the left yields (6.3). $\square$

Since up is even,

$$\int_0^1 t^{2n} \text{up}(t) dt = \frac{c_n}{2}. \quad (6.5)$$

6.2 Half-moments and a generating function

Define $d_0 = 1$ and, for $n \geq 1$,

$$d_n = n \int_0^1 t^{n-1} \text{up}(t) dt. \quad (6.6)$$

Introduce

$$f(z) = 1 + z \int_0^1 e^{zt} \text{up}(t) dt = \sum_{n=0}^{\infty} d_n \frac{z^n}{n!}. \quad (6.7)$$

The probability interpretation gives another form:

$$f(z) = e^{z/2} \widehat{\text{up}}\left(\frac{iz}{4\pi}\right). \quad (6.8)$$

Indeed, the right side is the moment-generating function of the random variable X in (3.6). Using $\text{up}(t) = 1 - F(t)$ on $[0, 1]$ and integration by parts gives the first expression in (6.7).

Proposition 6.2 (Refinement of the generating function). *The entire function f satisfies*

$$f(2z) = \frac{e^z - 1}{z} f(z), \quad (6.9)$$

with the quotient understood by continuity at $z = 0$. Therefore

$$d_0 = 1, \quad (n+1)(2^n - 1)d_n = \sum_{k=0}^{n-1} \binom{n+1}{k} d_k \quad (n \geq 1). \quad (6.10)$$

Proof. Substitute (6.8) into (3.4) at the imaginary argument $iz/(2\pi)$. The hyperbolic-sine simplification gives (6.9). Expanding both sides in exponential generating series and comparing coefficients gives (6.10). $\square$

The first values are

$$1, \quad \frac{1}{2}, \quad \frac{5}{18}, \quad \frac{1}{6}, \quad \frac{143}{1350}, \quad \frac{19}{270}, \quad \frac{1153}{23814}, \quad \frac{583}{17010}, \dots \quad (6.11)$$

Theorem 6.3 (Moment–endpoint identity). *For every $n \geq 1$,*

$$\int_0^1 t^{n-1} \text{up}(t) dt = (n-1)! 2^{\binom{n}{2}} F(2^{-n}). \quad (6.12)$$

Equivalently,

$$d_n = n! 2^{\binom{n}{2}} F(2^{-n}) = n! 2^{\binom{n}{2}} \text{up}(1 - 2^{-n}). \quad (6.13)$$

Proof. All derivatives of F vanish at 0. Taylor's theorem with integral remainder, applied at order $n-1$ and at $a = 2^{-n}$, gives

$$F(2^{-n}) = \frac{1}{(n-1)!} \int_0^{2^{-n}} (2^{-n} - t)^{n-1} F^{(n)}(t) dt.$$

Repeated differentiation of $F'(t) = 2F(2t)$ is valid throughout $0 \leq t \leq 2^{-n}$ and gives

$$F^{(n)}(t) = 2^{\binom{n+1}{2}} F(2^n t).$$

Set $u = 2^n t$. The powers of 2 simplify to $2^{-\binom{n}{2}}$, and hence

$$F(2^{-n}) = \frac{2^{-\binom{n}{2}}}{(n-1)!} \int_0^1 (1-u)^{n-1} F(u) du.$$

Finally $F(u) = \text{up}(u-1) = \text{up}(1-u)$, and the substitution $t = 1-u$ proves (6.12). Multiplication by n gives (6.13). $\square$

Proposition 6.4 (Half-moments from even moments). *For every $n \geq 0$,*

$$d_n = 2^{-n} \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k. \quad (6.14)$$

Proof. For $n \geq 1$, integrate (6.6) by parts. The boundary term vanishes because $\text{up}(1) = 0$, and the Rvachev equation on $(0, 1)$ gives

$$\begin{aligned} d_n &= - \int_0^1 t^n \text{up}'(t) dt = 2 \int_0^1 t^n \text{up}(2t - 1) dt \\ &= \int_{-1}^1 \left(\frac{x+1}{2} \right)^n \text{up}(x) dx. \end{aligned}$$

Expand $(x+1)^n$. Odd moments vanish by evenness, and the even moments are c_k . The case $n = 0$ is immediate. $\square$

7 A closed formula for every dyadic value

The rationality of dyadic values is often proved recursively. A more illuminating formula evaluates them directly from the even moments c_k and a finite Thue–Morse sum. It is valid not only on the unit interval but for the signed global extension.

Definition 7.1. For $n, a \in \mathbb{N}$, define

$$Q_n(a) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{a-1} \epsilon_h \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} (2a - 2h - 1)^{n-2k} c_k, \quad (7.1)$$

where the outer sum is empty when $a = 0$.

Theorem 7.2 (Exact dyadic formula). *For every $n, a \in \mathbb{N}$,*

$$\Theta\left(\frac{a}{2^n}\right) = Q_n(a). \quad (7.2)$$

Consequently, if $0 \leq a \leq 2^n$, then

$$F\left(\frac{a}{2^n}\right) = Q_n(a) \in \mathbb{Q}. \quad (7.3)$$

For an integer q ,

$$\text{up}\left(\frac{q}{2^n}\right) = \begin{cases} Q_n(2^n - |q|), & |q| \leq 2^n, \\ 0, & |q| > 2^n. \end{cases} \quad (7.4)$$

Proof. Because Θ and all its derivatives vanish on the negative half-line, repeated integration of its $(n+1)$ st derivative gives, for every real x ,

$$\Theta(x) = \frac{1}{n!} \int_{-\infty}^x (x-t)^n \Theta^{(n+1)}(t) dt. \quad (7.5)$$

By (5.4),

$$\Theta^{(n+1)}(t) = 2^{\binom{n+2}{2}} \Theta(2^{n+1}t).$$

Set $x = a/2^n$ and then $u = 2^{n+1}t$. The powers of 2 cancel to $2^{-\binom{n+1}{2}}$, yielding

$$\Theta\left(\frac{a}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!} \int_{-\infty}^{2a} (2a - u)^n \Theta(u) du. \quad (7.6)$$

Insert the locally finite bump decomposition (5.1). Up to the upper limit $2a$, exactly the complete bumps indexed by $0 \leq h < a$ occur:

$$\int_{-\infty}^{2a} (2a - u)^n \Theta(u) du = \sum_{h=0}^{a-1} \epsilon_h \int_{2h}^{2h+2} (2a - u)^n \text{up}(u - 2h - 1) du.$$

With $v = u - 2h - 1$, each integral becomes

$$\int_{-1}^1 (2a - 2h - 1 - v)^n \text{up}(v) dv.$$

Expand by the binomial theorem. The odd moments vanish by evenness, while

$$\int_{-1}^1 v^{2k} \text{up}(v) dv = c_k.$$

The result is precisely the double sum in (7.1).

On $[0, 1]$, $\Theta = F$, proving (7.3). Finally, evenness and the relation $\text{up}(t) = F(1 - |t|)$ for $|t| \leq 1$ give (7.4). $\square$

Remark 7.3 (Why the formula is finite). The finiteness has two independent sources. Only a translated bumps lie to the left of $2a$, and only even moments up to order n appear in the binomial expansion. No limiting operation and no numerical approximation remains. Since the recurrence (6.3) computes every c_k in $\mathbb{Q}$, the formula is an exact rational algorithm.

7.1 Examples

The first two nontrivial levels are

$$\begin{aligned} F(1/2) &= \frac{1}{2}, \\ F(1/4) &= \frac{5}{72}, & F(3/4) &= \frac{67}{72}, \\ F(1/8) &= \frac{1}{288}, & F(3/8) &= \frac{73}{288}, & F(5/8) &= \frac{215}{288}, & F(7/8) &= \frac{287}{288}. \end{aligned}$$

At denominator 16, selected values are

x	$F(x)$	x	$F(x)$
1/16	143/2073600	9/16	1295857/2073600
2/16	1/288	10/16	215/288
3/16	46657/2073600	11/16	1767743/2073600
4/16	5/72	12/16	67/72
5/16	305857/2073600	13/16	2026943/2073600
6/16	73/288	14/16	287/288
7/16	777743/2073600	15/16	2073457/2073600
8/16	1/2	16/16	1

The table exhibits both refinement invariance, for example $F(2/16) = F(1/8)$, and reflection $F(1 - x) = 1 - F(x)$.

7.2 Refinement invariance

A dyadic rational has many presentations. Correctness requires

$$Q_{n+1}(2a) = Q_n(a). \quad (7.7)$$

This identity follows immediately from [Theorem 7.2](#), but it can also be proved purely algebraically from the Thue–Morse relations (5.2) and the moment recurrence. In the formal development, both the closed expression and the executable evaluator are proved invariant under arbitrary repeated refinements $(a, n) \mapsto (2a, n+1)$; thus exact computation does not depend on first reducing the dyadic fraction.

8 A fast bit-recursive exact evaluator

The closed formula sums over every $h < a$. It is excellent for theory but can be wasteful when a has only a few nonzero binary digits. A terminating Taylor identity removes one highest set bit at each step.

8.1 Taylor data at inverse powers of two

For $0 \leq k \leq r$, repeated differentiation on the left half and (6.13) give

$$F^{(k)}(2^{-r}) = 2^{\binom{k+1}{2}} F(2^{-(r-k)}) = 2^{\binom{k+1}{2} - \binom{r-k}{2}} \frac{d_{r-k}}{(r-k)!}. \quad (8.1)$$

All derivatives of order greater than r vanish at 2^{-r} .

Theorem 8.1 (Terminating Taylor recursion). *Let $r \geq 1$ and $0 \leq y < 2^{-r}$. Then*

$$F(2^{-r} + y) = -F(y) + \sum_{k=0}^r 2^{\binom{k+1}{2} - \binom{r-k}{2}} \frac{d_{r-k}}{(r-k)!} \frac{y^k}{k!}. \quad (8.2)$$

More generally, the same formula holds for the signed global extension on the corresponding dyadic block, with the empty sum convention in the ranges where the Taylor center is an even integer.

Proof. Apply Taylor's theorem with integral remainder of order r at 2^{-r} . The polynomial part is (8.1). It remains to identify the remainder. Using $F = \Theta$ on the unit interval and (5.4), the $(r+1)$ st derivative on the short interval to the right of 2^{-r} is a rescaled copy of the next signed bump. More explicitly, after the change of variables dictated by $\Theta^{(r+1)}(t) = 2^{\binom{r+2}{2}} \Theta(2^{r+1}t)$, the integral remainder is

$$-\frac{1}{r!} \int_0^y (y-v)^r F^{(r+1)}(v) dv.$$

Taylor's integral formula at the flat point 0 identifies the last integral with $-F(y)$. This is the analytic origin of the minus sign in (8.2). $\square$

8.2 Precomputation at inverse powers

Let

$$V_j = F(2^{-j}), \quad V_0 = 1.$$

The d -recurrence or (6.13) computes these values. A direct triangular recurrence used by the exact evaluator is

$$V_{n+1} = \frac{1}{2^{n+1} - 1} \sum_{k=0}^n \frac{V_k}{2^{\binom{n+1}{2} - \binom{k}{2}} (n+2-k)!}. \quad (8.3)$$

For example, $V_1 = 1/2$, $V_2 = 5/72$, $V_3 = 1/288$, and $V_4 = 143/2073600$.

8.3 The binary algorithm

Suppose $0 < a < 2^n$ and let 2^ℓ be the highest power of two not exceeding a . Put

$$r = n - \ell, \quad b = a - 2^\ell, \quad y = \frac{b}{2^n}.$$

Then $a/2^n = 2^{-r} + y$ and $0 \leq y < 2^{-r}$, so (8.2) reduces $F(a/2^n)$ to a known Taylor polynomial minus $F(b/2^n)$. The numerator b is obtained by deleting the highest set bit of a .

Algorithm 8.2 (Exact bounded dyadic evaluation). To compute $F(a/2^n)$ exactly:

1. Return 0 if $a \leq 0$ and 1 if $a \geq 2^n$.
2. Precompute $V_0, \dots, V_n$ by (8.3), or equivalently precompute $d_0, \dots, d_n$ by (6.10).
3. If $0 < a < 2^n$, remove the highest set bit as above, evaluate the finite Taylor polynomial in (8.2) by Horner's rule, and subtract the recursive value at $b/2^n$.

The recursion depth is exactly the number of set bits of a .

Thus a numerator such as $a = 2^{100} + 2^7 + 1$ requires only three recursive Taylor steps, whereas the closed sum (7.1) contains roughly 2^{100} outer terms if used naively.

8.4 Bounded, global, and rational interfaces

For the bounded function, signed dyadic arguments are clamped according to Definition 2.1. The signed global extension has a different right-hand behavior. Let $s = 2^n$, let $a > 0$, and write

$$a = 2sB + R, \quad 0 \leq R < 2s.$$

Set $C = R$ if $R \leq s$ and $C = 2s - R$ otherwise. Then

$$\Theta(a/2^n) = \epsilon_B F(C/2^n). \quad (8.4)$$

This is the triangular reduction modulo 2, followed by the Thue–Morse sign of the block. It gives a total exact evaluator for Θ on all dyadic arguments.

A reduced rational p/q is dyadic exactly when q is a power of two. Hence an exact rational interface first checks this denominator condition, returns “not dyadic” otherwise, and invokes the signed-integer dyadic evaluator when the condition holds. No floating-point recognition of powers of two is needed.

9 The finite q -binomial–Thue–Morse formula

The formula of [Theorem 7.2](#) uses ordinary binomial coefficients and moments. A different expression, motivated by a conjectured symbolic formula, uses Gaussian binomial coefficients at $q = 1/2$ and power sums with Thue–Morse signs. Its proof in the formal development is stronger than the original statement: an arbitrary common translation may be inserted in all inner powers without changing the answer.

For $q \in \mathbb{Q}$, define

$$(q; q)_n = \prod_{j=1}^n (1 - q^j), \quad \left[\begin{matrix} n \\ k \end{matrix} \right]_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}} \quad (0 \leq k \leq n). \quad (9.1)$$

For $c \in \mathbb{Q}$ and natural m, k, p , put

$$S_{m,k,p}(c) = \sum_{j=0}^{m2^k-1} \epsilon_j (j - m2^k + c)^p. \quad (9.2)$$

Theorem 9.1 (q -binomial formula). *For all $m, n \in \mathbb{N}$ and every common shift $c \in \mathbb{Q}$,*

$$\Theta \left(\frac{m}{2^n} \right) = \frac{1}{2^{n^2} (1/2; 1/2)_n} \sum_{k=0}^n \frac{\left[\begin{matrix} n \\ k \end{matrix} \right]_{1/2}}{2^{k(k-1)} (n+k)!} S_{m,k,n+k}(c). \quad (9.3)$$

The right side is independent of c . If $m \leq 2^n$, it is the bounded value $F(m/2^n)$. In particular, $m = 1$ gives a finite formula for $F(2^{-n})$.

Proof. We give the algebraic mechanism in several steps.

Step 1: exponential generating functions for the inner sums. For fixed m and k ,

$$\sum_{p \geq 0} S_{m,k,p}(c) \frac{z^p}{p!} = e^{(c-m2^k)z} \sum_{j=0}^{m2^k-1} \epsilon_j e^{jz}. \quad (9.4)$$

Splitting $j = h2^k + r$ and using $\epsilon_{h2^k+r} = \epsilon_h \epsilon_r$ gives the block factorization

$$\sum_{j=0}^{m2^k-1} \epsilon_j e^{jz} = \left(\sum_{h=0}^{m-1} \epsilon_h e^{h2^k z} \right) \prod_{r=0}^{k-1} (1 - e^{2^r z}). \quad (9.5)$$

The product on the right has a zero of exact order k at $z = 0$; this is the classical vanishing of Thue–Morse power sums below degree k .

Step 2: remove the forced zero. Divide the product in (9.5) by z^k . Its constant term is $(-1)^k 2^{\binom{k}{2}}$. After this normalization, the coefficient of z^n is exactly $S_{m,k,n+k}(c)/(n+k)!$ up to the displayed power of 2 and sign. Thus the outer sum in (9.3) is a weighted coefficient extraction from a family of refinement-factor series indexed by k .

Step 3: the Gaussian-binomial annihilator. The finite q -binomial theorem

$$\sum_{k=0}^n (-1)^k q^{\binom{k}{2}} \left[\begin{matrix} n \\ k \end{matrix} \right]_q X^k = \prod_{j=0}^{n-1} (1 - Xq^j) \quad (9.6)$$

shows that the weights at the nodes 2^{-k} annihilate every polynomial in that node of degree less than n and evaluate the degree- n term by

$$2^{\binom{n+1}{2}} (1/2; 1/2)_n. \quad (9.7)$$

Applied coefficientwise to the normalized refinement factors, all lower coefficients disappear. The surviving coefficient is the n th coefficient of the product of two series: the finite Thue–Morse prefix series associated with m , and the recurrence series

$$\sum_{r \geq 0} d_r \frac{z^r}{r!},$$

which satisfies the refinement equation (6.9).

Step 4: identify the surviving coefficient. The convolution coefficient just described is the exact global dyadic value $\Theta(m/2^n)$ in the recurrence normalization. The scalar in (9.7), together with the powers extracted in Step 2, simplifies to the denominator $2^{n^2}(1/2; 1/2)_n$ in (9.3).

Step 5: translation invariance. Replacing $c = 0$ by a common c multiplies every inner generating function by e^{cz} . In the coefficient convolution, the terms involving a positive power of z from e^{cz} are paired with weighted coefficients of degree strictly below n , and these are annihilated by Step 3. Only the constant term 1 survives. Hence the complete expression is independent of c . This proves the theorem. $\square$

Remark 9.2. The source formula uses $c = 1/2$, giving powers $(j - m2^k + 1/2)^{n+k}$. The centered choice $c = 0$ is often simpler for parity arguments. The theorem says that this is not a heuristic simplification but an exact invariance of the full outer sum. The formal proof also transports the identity from $\mathbb{Q}$ to $\mathbb{R}$ and $\mathbb{C}$ by coercion, after the rational identity has been established.

10 Arithmetic sequences, denominators, and parity

10.1 Integer normalizations of the moments

Let $(2n + 1)!! = 1 \cdot 3 \cdots (2n + 1)$. Define

$$\mathcal{F}_n = c_n(2n + 1)!! \prod_{\nu=1}^n (2^{2\nu} - 1). \quad (10.1)$$

We use the calligraphic symbol $\mathcal{F}_n$ to avoid confusion with the function F .

Proposition 10.1. *Every $\mathcal{F}_n$ is a positive integer. More precisely, $\mathcal{F}_0 = 1$ and, for $n \geq 1$,*

$$\mathcal{F}_n = \sum_{k=0}^{n-1} \mathcal{F}_k \binom{2n+1}{2k} \frac{(2n-1)!!}{(2k+1)!!} \prod_{\nu=k+1}^{n-1} (2^{2\nu} - 1). \quad (10.2)$$

Proof. Substitute (10.1) into the moment recurrence (6.3) and clear the common factors. Every factor on the right of (10.2) is an integer; induction proves the claim. $\square$

Similarly define G_n by

$$d_n = \frac{G_n}{(n+1)! \prod_{j=1}^n (2^j - 1)}. \quad (10.3)$$

Then $G_0 = 1$ and

$$G_n = \sum_{k=0}^{n-1} G_k \binom{n+1}{k} \frac{n!}{(k+1)!} \prod_{j=k+1}^{n-1} (2^j - 1), \quad (10.4)$$

so all G_n are positive integers. The first terms are

$$1, 1, 5, 84, 4004, 494760, 150120600, 107969547840, \dots$$

Combining (10.3) with (6.13) gives an exact integer-normalized formula for the inverse-power values:

$$F(2^{-n}) = \frac{2^{-\binom{n}{2}} G_n}{n!(n+1)! \prod_{j=1}^n (2^j - 1)}. \quad (10.5)$$

10.2 Reshetnikov's integers

For $n \geq 1$, define

$$R_n = 2^{\binom{n-1}{2}} (2n)! F(2^{-n}) \prod_{j=1}^{\lfloor n/2 \rfloor} (2^{2j} - 1). \quad (10.6)$$

The positive exponent in (10.6) is essential; a sign typo in one summary of the formula reverses it.

Using (6.13) and $(2n)! = 2^n n! (2n-1)!!$ gives

$$R_n = 2d_n (2n-1)!! \prod_{j=1}^{\lfloor n/2 \rfloor} (2^{2j} - 1). \quad (10.7)$$

For odd indices,

$$R_{2n+1} = \mathcal{F}_n \frac{(4n+1)!!}{(2n-1)!!}. \quad (10.8)$$

For even indices,

$$R_{2n} = \sum_{k=0}^n \frac{2\mathcal{F}_k}{2^{2n}} \binom{2n}{2k} \frac{(4n-1)!!}{(2k+1)!!} \prod_{\ell=k+1}^n (2^{2\ell} - 1). \quad (10.9)$$

The latter formula shows first that the only possible denominator is a power of two.

Theorem 10.2 (Integrality of R_n). *Every R_n is a positive integer.*

Proof. The odd case follows at once from (10.8). By (10.7), $\nu_2(R_n) = \nu_2(2d_n)$ because the remaining factors are odd. We prove $\nu_2(2d_n) \geq 0$ by induction. Odd indices are already covered by (10.8), and $2d_0 = 2$. For an even index $2n$, multiply (6.10) by 2 to obtain

$$(2n+1)(2^{2n} - 1)2d_{2n} = \sum_{k=0}^{2n-1} \binom{2n+1}{k} 2d_k. \quad (10.10)$$

The left prefactor is odd. By induction every summand on the right is 2-integral, hence so is $2d_{2n}$. Formula (10.9) says there are no odd-prime denominators, so R_{2n} is an integer. Positivity follows from $F(2^{-n}) > 0$. $\square$

10.3 Denominator bounds for all dyadic values

Let D_n be the least common multiple of the reduced denominators of $\text{up}(a/2^n)$ for odd integers a with $|a| < 2^n$. Refinement invariance shows that this captures the genuinely new denominator at level n .

Theorem 10.3 (Uniform denominator bound). *For $n \geq 1$,*

$$D_n \mid n! 2^{\binom{n+1}{2}} (2\lfloor n/2 \rfloor + 1)!! \prod_{k=1}^{\lfloor n/2 \rfloor} (2^{2k} - 1). \quad (10.11)$$

Moreover, D_n is a multiple of the denominator of

$$\frac{d_n}{2^{\binom{n}{2}} n!} = F(2^{-n}). \quad (10.12)$$

Proof. After multiplication by $n!2^{\binom{n+1}{2}}$, the closed formula (7.1) is an integral linear combination of $c_0, \dots, c_{\lfloor n/2 \rfloor}$. Formula (10.1) shows that the common odd factor

$$(2\lfloor n/2 \rfloor + 1)!! \prod_{k=1}^{\lfloor n/2 \rfloor} (2^{2k} - 1)$$

clears all their denominators. This proves (10.11). The second claim follows because the inverse-power value $\text{up}(1 - 2^{-n}) = F(2^{-n})$ is one of the values whose denominator divides D_n . $\square$

The following striking pattern is supported by computation and is recorded as a conjecture, not as a theorem.

Conjecture 10.4 (Denominator pattern). *Put $A_n = 2^{-\binom{n}{2}} D_n$. Then*

- (a) $A_{2n} = A_{2n+1}$ for $n \geq 1$;
- (b) if $K_n = A_{2n-1}$, then $2(2n-1)!$ divides K_n ;
- (c) $H_n = K_n/[2(2n-1)!]$ is an odd integer.

10.4 Lucas parity and exact 2-adic valuations

Lucas's theorem modulo 2 says that $\binom{N}{K}$ is odd exactly when every binary digit of K is bounded by the corresponding digit of N . Since the binary expansion of $2n+1$ is the expansion of n followed by a final 1, we obtain:

Proposition 10.5. *For $n \geq 1$, exactly $2^{\text{wt}(n)}$ of the numbers $\binom{2n+1}{2k}$, $0 \leq k \leq n$, are odd. Exactly $2^{\text{wt}(n)+1}$ of all the numbers $\binom{2n+1}{k}$ are odd.*

Theorem 10.6 (Oddness theorem). *For every $n \geq 0$, $\mathcal{F}_n$ is odd. For every $n \geq 1$,*

$$\nu_2(2d_n) = 0. \tag{10.13}$$

Consequently every R_n is odd, and

$$\nu_2(\text{up}(1 - 2^{-n})) = -\binom{n}{2} - 1 - \nu_2(n!). \tag{10.14}$$

Proof. Use (10.2) and induction. All factors other than the binomial coefficient are odd. Among the $k = 0, \dots, n-1$ terms, the number of odd binomial coefficients is $2^{\text{wt}(n)} - 1$, because the omitted $k = n$ coefficient is odd. This number is odd for $n \geq 1$, so the sum is odd.

For odd indices, (10.8) and (10.7) imply $\nu_2(2d_{2m+1}) = 0$. For even indices, use the induction equation (10.10) after multiplying by a sufficiently large common odd denominator. By the induction hypothesis the terms with $k \geq 1$ are odd, while the $k = 0$ term is even. Proposition 10.5 implies that an odd number of the remaining summands are odd, so the total is odd. Hence (10.13). Equation (10.7) then makes R_n odd. Finally take 2-adic valuations in (6.13). $\square$

10.5 Bernoulli recurrences

Let B_j be the Bernoulli numbers with $B_1 = -1/2$. The reciprocal refinement factors

$$\frac{\pi z}{\sin \pi z}, \quad \frac{z}{e^z - 1}$$

give alternative recurrences.

Proposition 10.7. *For $n \geq 1$,*

$$c_n = \frac{1}{2^{2n} - 1} \sum_{k=1}^n 2^{2n-2k} (2^{2k} - 2) \binom{2n}{2k} B_{2k} c_{n-k}, \quad (10.15)$$

and

$$d_n = \frac{n 2^{n-2}}{2^n - 1} d_{n-1} - \frac{1}{2^n - 1} \sum_{k=1}^{\lfloor n/2 \rfloor} \binom{n}{2k} 2^{n-2k} B_{2k} d_{n-2k}. \quad (10.16)$$

Proof. Rewrite the Fourier and generating-function refinements as

$$\frac{\pi z}{\sin \pi z} \widehat{\text{up}}(z) = \widehat{\text{up}}(z/2), \quad \frac{z}{e^z - 1} f(2z) = f(z).$$

Insert the standard Bernoulli expansions of the two reciprocal factors and compare coefficients. $\square$

11 The Fourier–Legendre expansion

Because up is even, its Legendre expansion contains only even degrees. Let P_m be the Legendre polynomial normalized by $P_m(1) = 1$, and put

$$u_n = \frac{4n+1}{2} \int_{-1}^1 \text{up}(x) P_{2n}(x) dx. \quad (11.1)$$

Theorem 11.1 (Exact Legendre coefficients). *For every $n \geq 0$,*

$$u_n = 4^{-n} (4n+1) \sum_{k=0}^n (-1)^{n+k} \binom{2n}{n+k} \binom{2n+2k}{2n} \cdot (2k)! 2^{\binom{2k+1}{2}} F(2^{-2k-1}). \quad (11.2)$$

Moreover,

$$\text{up}(x) = \sum_{n=0}^{\infty} u_n P_{2n}(x) \quad (11.3)$$

converges absolutely and uniformly on $[-1, 1]$, including both endpoints.

Proof. The explicit monomial expansion

$$P_{2n}(x) = 4^{-n} \sum_{k=0}^n (-1)^{n+k} \binom{2n}{n+k} \binom{2n+2k}{2n} x^{2k} \quad (11.4)$$

reduces (11.1) to the even moments c_k . From (6.14) in odd degree, or directly from (6.6) and evenness,

$$d_{2k+1} = \frac{2k+1}{2} c_k.$$

Combining this with (6.13) gives

$$\frac{c_k}{2} = (2k)! 2^{\binom{2k+1}{2}} F(2^{-2k-1}). \quad (11.5)$$

Substitution into (11.1) proves (11.2).

For convergence, use the self-adjoint Legendre operator

$$\mathcal{L}g = -\frac{d}{dx} \left((1-x^2)g'(x) \right), \quad \mathcal{L}P_m = m(m+1)P_m.$$

All derivatives of up vanish at ± 1 , so repeated integration by parts has no boundary terms. For every $r \geq 1$,

$$\int_{-1}^1 \text{up } P_m = \frac{1}{[m(m+1)]^r} \int_{-1}^1 \mathcal{L}^r \text{up } P_m.$$

Since $|P_m(x)| \leq 1$ on $[-1, 1]$ and $\mathcal{L}^r \text{up}$ is continuous, the ordinary Legendre coefficient is $O(m^{1-2r})$. Taking $r \geq 2$ makes the coefficient series absolutely summable. The Weierstrass test yields absolute uniform convergence. Orthogonality identifies its sum with up in L^2 , and uniform convergence plus continuity upgrades the identity to every point. $\square$

Remark 11.2. Formula (11.2) is an unusual bridge between a global orthogonal-polynomial expansion and endpoint values at the sparse sequence $2^{-1}, 2^{-3}, 2^{-5}, \dots$. Every coefficient is rational, although the series represents a transcendental-looking smooth bump.

Endpoint asymptotics

12 The negative Laplace product and dyadic renormalization

The endpoint $x \downarrow 0$ is governed by the Laplace transform of the random series X . Put, for $s > 0$,

$$G(s) = f(-s) = \mathbb{E}e^{-sX}. \quad (12.1)$$

The refinement equation (6.9) gives

$$G(s) = \prod_{j=1}^{\infty} \frac{1 - e^{-s/2^j}}{s/2^j}. \quad (12.2)$$

This is the basic product whose logarithm must be analyzed sharply.

12.1 The exact Laplace identity

Proposition 12.1. *For $s > 0$,*

$$\int_0^{\infty} e^{-sx} F(x) dx = \frac{G(s)}{s}. \quad (12.3)$$

Here $F(x) = 1$ for $x \geq 1$, as in the bounded convention.

Proof. The distribution function of X is F . Integration by parts on $[0, 1]$ gives

$$\int_0^1 e^{-sx} F(x) dx = -\frac{e^{-s}}{s} + \frac{1}{s} \int_0^1 e^{-sx} dF(x) = -\frac{e^{-s}}{s} + \frac{G(s)}{s}.$$

The constant tail contributes $\int_1^{\infty} e^{-sx} dx = e^{-s}/s$, and the two boundary terms cancel. $\square$

For every $r > 0$, Bromwich inversion therefore has the form

$$F(x) = \frac{1}{2\pi i} \int_{r-i\infty}^{r+i\infty} e^{sx} \frac{G(s)}{s} ds, \quad x > 0. \quad (12.4)$$

The compact support of the underlying probability law and the rapid product bounds justify the inversion and the contour manipulations used below.

12.2 An exact periodic decomposition

Define

$$K(u) = \log \left(\frac{1 - e^{-u}}{u} \right), \quad \mathcal{L}(s) = \log G(s) = \sum_{j=1}^{\infty} K(s/2^j). \quad (12.5)$$

The series is locally uniformly convergent for $s > 0$. It obeys

$$\mathcal{L}(2s) - \mathcal{L}(s) = K(s). \quad (12.6)$$

Let

$$H(s) = \mathcal{L}(s) + \frac{(\log s)^2}{2L} - \frac{\log s}{2}. \quad (12.7)$$

A direct calculation using $K(s) + \log s = \log(1 - e^{-s})$ gives

$$H(2s) - H(s) = \log(1 - e^{-s}). \quad (12.8)$$

The defect is removed by the forward tail

$$E(s) = - \sum_{m=0}^{\infty} \log(1 - e^{-2^m s}). \quad (12.9)$$

Indeed,

$$E(2s) - E(s) = \log(1 - e^{-s}). \quad (12.10)$$

Consequently $H(s) - E(s)$ is invariant under $s \mapsto 2s$.

Theorem 12.2 (Exact dyadic decomposition). *There is a smooth one-periodic function $R : \mathbb{R} \rightarrow \mathbb{R}$ such that, for every $s > 0$,*

$$\mathcal{L}(s) = -\frac{(\log s)^2}{2L} + \frac{\log s}{2} + R(\log_2 s) + E(s). \quad (12.11)$$

The tail is positive and satisfies

$$0 \leq E(s) \leq \frac{e^{-s}}{(1 - e^{-s})^2}. \quad (12.12)$$

In particular, $E(s)$ is exponentially small as $s \rightarrow \infty$ and is smaller than every negative power of $\log s$.

Proof. Define $R(\log_2 s) = H(s) - E(s)$. Equations (12.8) and (12.10) make this definition invariant under replacing s by $2s$, hence R is one-periodic. Smoothness follows from locally uniform differentiation of the defining series.

For the bound, use $-\log(1 - y) \leq y/(1 - y)$ and $2^m \geq m + 1$:

$$E(s) \leq \sum_{m \geq 0} \frac{e^{-2^m s}}{1 - e^{-2^m s}} \leq \frac{1}{1 - e^{-s}} \sum_{m \geq 0} e^{-(m+1)s} = \frac{e^{-s}}{(1 - e^{-s})^2}.$$

Rearranging the definition of R proves (12.11). $\square$

The term $R(\log_2 s)$ is not an artifact of the proof. It is the exact record of the discrete dyadic scale in the infinite product. Its mean and all Fourier coefficients can be computed explicitly.

13 The Gamma–zeta periodic fluctuation

Set

$$A_0 = \frac{6\gamma^2 + 12\gamma_1 - \pi^2}{12} = \frac{\gamma^2}{2} + \gamma_1 - \frac{\pi^2}{12}. \quad (13.1)$$

Theorem 13.1 (Mean and Fourier coefficients). *The mean of the periodic function in (12.11) is*

$$\bar{R} = \int_0^1 R(u) du = \frac{A_0}{L} - \frac{L}{12}. \quad (13.2)$$

Let

$$\Psi(u) = R(u) - \bar{R}. \quad (13.3)$$

Then Ψ is real-valued, one-periodic, and has the absolutely convergent Fourier series

$$\Psi(u) = \sum_{k \in \mathbb{Z} \setminus \{0\}} \hat{\Psi}(k) e^{2\pi i k u}, \quad (13.4)$$

where

$$\hat{\Psi}(k) = -\frac{1}{L} \Gamma\left(-\frac{2\pi i k}{L}\right) \zeta\left(1 - \frac{2\pi i k}{L}\right) \quad (k \neq 0). \quad (13.5)$$

Every nonzero Fourier coefficient is nonzero. In particular, Ψ is not constant and cannot be absorbed into an error term tending to zero.

Proof. The Mellin transform of K in the fundamental strip $-1 < \Re z < 0$ is

$$\int_0^\infty K(u) u^{z-1} du = -\Gamma(z) \zeta(1+z). \quad (13.6)$$

One obtains this by integration by parts from $K'(u) = (e^u - 1)^{-1} - u^{-1}$ and then by analytic continuation of the classical Mellin integral for $(e^u - 1)^{-1}$.

Mellin inversion of the harmonic sum $\sum_{j \geq 1} K(s/2^j)$ introduces the geometric factor $2^z/(1 - 2^z)$. Moving the contour across the imaginary-axis poles $z = 2\pi i k/L$ yields the Fourier modes. With the Fourier convention in (13.4), the residue at the pole indexed by $-k$ is exactly (13.5).

At $z = 0$ there is a third-order pole. The Laurent expansions

$$\begin{aligned} \Gamma(z) &= \frac{1}{z} - \gamma + \left(\frac{\gamma^2}{2} + \frac{\pi^2}{12}\right) z + O(z^2), \\ \zeta(1+z) &= \frac{1}{z} + \gamma - \gamma_1 z + O(z^2), \quad \frac{2^z}{1-2^z} = -\frac{1}{Lz} - \frac{1}{2} - \frac{Lz}{12} + O(z^3) \end{aligned}$$

give the quadratic and linear logarithmic terms already extracted in (12.11); the remaining constant residue is $A_0/L - L/12$, proving (13.2).

Stirling's formula on vertical lines makes the Gamma factor exponentially small in $|k|$, while $\zeta(1+it)$ has at most subexponential growth. Hence the Fourier series and all its termwise derivatives converge absolutely. The Gamma function has no zeros, and the classical zero-free line $\zeta(1+it) \neq 0$ for $t \neq 0$ shows that every coefficient in (13.5) is nonzero. $\square$

Numerically,

$$\hat{\Psi}(1) \approx (7.5586467408 - 7.4700963646i) \times 10^{-7}, \quad (13.7)$$

$$|\hat{\Psi}(1)| \approx 1.0627110626 \times 10^{-6}. \quad (13.8)$$

The second harmonic has magnitude about 6.69×10^{-13} and subsequent ones are dramatically smaller. Thus the oscillation is visually tiny, but its amplitude does not decay as $x \downarrow 0$; asymptotically, that is what matters.

14 The exact lower-Lambert phase

The saddle scale is not simply $\log_2(1/x)$. It is the large solution of a Lambert equation.

Definition 14.1. For $0 < x < 1/(eL)$, define

$$\lambda(x) = -\frac{1}{L}W_{-1}(-Lx), \quad (14.1)$$

where W_{-1} is the lower real branch of Lambert's W -function.

The defining identity $We^W = z$ gives

$$\lambda(x)2^{-\lambda(x)} = x. \quad (14.2)$$

Thus, with

$$r(x) = 2^{\lambda(x)}, \quad (14.3)$$

we have the exact saddle relation

$$r(x)x = \lambda(x), \quad r(x) = \frac{\lambda(x)}{x}. \quad (14.4)$$

The lower branch is essential: it selects the solution $\lambda > 1/L$ and hence $\lambda(x) \rightarrow \infty$ as $x \downarrow 0$.

If $x = 2^{-t}$, then (14.2) is equivalent to the fixed point equation

$$\lambda - \frac{\log \lambda}{L} = t. \quad (14.5)$$

In particular,

$$\lambda(t) \sim t, \quad \lambda(t) - t \sim \frac{\log t}{L}. \quad (14.6)$$

This distinction is large enough to affect several lower-order terms in $\log F(2^{-t})$.

15 The sharp endpoint main term

Define the constant

$$C_* = \frac{A_0}{L} - \frac{7L}{12} - \frac{1}{2} \log \pi = \frac{6\gamma^2 + 12\gamma_1 - \pi^2}{12L} - \frac{7L}{12} - \frac{1}{2} \log \pi. \quad (15.1)$$

For sufficiently small $x > 0$, set

$$\begin{aligned} M(x) = & -\frac{L}{2}\lambda(x)^2 + \left(1 + \frac{L}{2}\right)\lambda(x) - \frac{1}{2}\log \lambda(x) \\ & + C_* + \Psi(\lambda(x)). \end{aligned} \quad (15.2)$$

Theorem 15.1 (Sharp endpoint asymptotic). *As $x \downarrow 0$,*

$$\log F(x) = M(x) + O\left(\frac{1}{\lambda(x)}\right) = M(x) + O\left(\frac{1}{-\log x}\right). \quad (15.3)$$

More precisely,

$$F(x) = e^{M(x)} \left(1 + O\left(\frac{1}{\lambda(x)}\right)\right). \quad (15.4)$$

The periodic function in (15.2) is evaluated at the exact phase (14.1).

Proof. We describe the saddle argument in enough detail to locate every term in (15.2). Start from Bromwich inversion (12.4), choose $r = 2^\lambda$ with $rx = \lambda$, and parametrize the vertical line by

$$s = r(1 + iy), \quad y \in \mathbb{R}.$$

Since $ds/s = i dy/(1 + iy)$,

$$F(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\exp\{\lambda(1 + iy) + \mathcal{L}(r(1 + iy))\}}{1 + iy} dy. \quad (15.5)$$

The real decomposition (12.11) extends analytically to the part of the right half-plane used by the central contour. Put $\delta(y) = \text{Log}(1 + iy)$. Then

$$\text{Log } s = L\lambda + \delta(y), \quad \log_2 s = \lambda + \frac{\delta(y)}{L}.$$

Ignoring for the moment the exponentially small forward tail, the logarithm of the integrand, including the factor $(1 + iy)^{-1}$, is

$$\begin{aligned} \mathcal{E}_\lambda(y) = \lambda(1 + iy) - \frac{(L\lambda + \delta(y))^2}{2L} + \frac{L\lambda + \delta(y)}{2} \\ + \bar{R} + \Psi\left(\lambda + \frac{\delta(y)}{L}\right) - \delta(y). \end{aligned} \quad (15.6)$$

At $y = 0$ this equals

$$-\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda + \bar{R} + \Psi(\lambda). \quad (15.7)$$

The choice $rx = \lambda$ cancels the linear term in y . Since

$$\delta(y) = iy + \frac{y^2}{2} - \frac{iy^3}{3} - \frac{y^4}{4} + O(y^5),$$

the real part of the exponent begins with $-\lambda y^2/2$. On the central scale $y = v/\sqrt{\lambda}$, the normalized integral therefore tends to

$$\frac{1}{2\pi\sqrt{\lambda}} \int_{-\infty}^{\infty} e^{-v^2/2} dv = \frac{1}{\sqrt{2\pi\lambda}}.$$

Taking logarithms contributes $-\frac{1}{2}\log \lambda - \frac{1}{2}\log(2\pi)$. Since

$$\bar{R} - \frac{1}{2}\log(2\pi) = \frac{A_0}{L} - \frac{L}{12} - \frac{L}{2} - \frac{1}{2}\log \pi = C_*,$$

we recover exactly (15.2).

For completeness, the error analysis has three parts. On a shrinking central arc, Taylor expansion of (15.6), dominated by a fixed Gaussian, gives a relative error $O(1/\lambda)$. On the adjacent arc, the real part is bounded above by $-c\lambda y^2$. On the remaining minor arc, blocks of factors in the product (12.2) give uniform geometric suppression; the resulting integral is exponentially smaller than the central mass. Finally, (12.12) at $r = 2^\lambda$ shows that the forward tail is exponentially flat in λ and does not affect any inverse-power term. These estimates justify contour truncation, termwise expansion, and the passage from the relative estimate (15.4) to (15.3). $\square$

15.1 The leading quadratic law

Corollary 15.2. As $x \downarrow 0$,

$$\frac{\log F(x)}{(\log x)^2} \longrightarrow -\frac{1}{2 \log 2}. \quad (15.8)$$

Equivalently, as $t \rightarrow \infty$,

$$\frac{\log F(2^{-t})}{t^2} \longrightarrow -\frac{\log 2}{2}. \quad (15.9)$$

Consequently, for every $A, \alpha > 0$,

$$F(x) = o(x^A), \quad e^{-x^{-\alpha}} = o(F(x)). \quad (15.10)$$

Proof. The phase satisfies $\lambda \sim -\log x/L$, and every other term in (15.2) is $o(\lambda^2)$. This gives (15.8) and (15.9). The comparisons in (15.10) follow by comparing the quadratic logarithmic scale with $\log x$ and with $-x^{-\alpha}$, respectively. $\square$

16 Corrected explicit forms and the failure of the nonperiodic formula

16.1 The compact Lambert- W form

Let $W = W_{-1}(-Lx)$. Since $W = -L\lambda(x)$ and $\log x = \log \lambda - L\lambda$, the main term (15.2) is exactly

$$M(x) = -\frac{7L}{12} - \frac{1}{2} \log(\pi x) + \frac{A_0 - W - W^2/2}{L} + \Psi\left(-\frac{W}{L}\right). \quad (16.1)$$

This is the cleanest closed expression for the corrected leading asymptotic. Theorem 15.1 says that

$$\log F(x) = \text{right side of (16.1)} + O\left(\frac{1}{-\log x}\right). \quad (16.2)$$

16.2 An elementary logarithmic expansion

To compare with formulas written without Lambert W , set

$$\ell = \log x < 0, \quad m = \log(-\ell), \quad a = \log L. \quad (16.3)$$

Define

$$\begin{aligned} \mathcal{E}_0(x) = & -\frac{\ell^2}{2L} + \frac{\ell m}{L} - \left(\frac{1}{2} + \frac{1+a}{L}\right)\ell - \frac{m^2}{2L} + \frac{am}{L} \\ & + \frac{6\gamma^2 + 12\gamma_1 - \pi^2 - 6a^2}{12L} - \frac{7L}{12} - \frac{1}{2} \log \pi. \end{aligned} \quad (16.4)$$

Expansion of the exact phase gives

$$\log F(x) = \mathcal{E}_0(x) + \Psi(\lambda(x)) + O\left(\frac{m^2}{-\ell}\right). \quad (16.5)$$

Keeping the first inverse-logarithmic phase term gives

$$\mathcal{E}_1(x) = \mathcal{E}_0(x) + \frac{(m-a)^2}{2L\ell} \quad (16.6)$$

and

$$\log F(x) = \mathcal{E}_1(x) + \Psi(\lambda(x)) + O\left(\frac{1}{-\ell}\right). \quad (16.7)$$

The sign in the last term is encoded by $\ell < 0$.

16.3 Dyadic logarithmic scale

For $x = 2^{-t}$, write $u = \log t$. Then the exact phase satisfies (14.5), and the main term expands as

$$\begin{aligned} M(2^{-t}) = & -\frac{L}{2}t^2 - tu + \left(1 + \frac{L}{2}\right)t - \frac{u^2}{2L} + C_* + \Psi(\lambda(t)) \\ & - \frac{u^2}{2L^2t} + O\left(\frac{u^3}{t^2}\right). \end{aligned} \quad (16.8)$$

Combining this with Theorem 15.1 gives a convenient corrected formula with error $O(1/t)$.

16.4 Why the periodic term cannot be omitted

Let

$$M_0(x) = M(x) - \Psi(\lambda(x)). \quad (16.9)$$

Then Theorem 15.1 gives

$$\log F(x) - M_0(x) = \Psi(\lambda(x)) + o(1). \quad (16.10)$$

Choose any $u \in [0, 1)$ and the sequence

$$\lambda_N = N + u, \quad x_N = \lambda_N 2^{-\lambda_N}. \quad (16.11)$$

Then $x_N \downarrow 0$ and the exact phase of x_N is λ_N , so

$$\log F(x_N) - M_0(x_N) \longrightarrow \Psi(u). \quad (16.12)$$

Because Ψ is nonconstant, these subsequential limits are not all equal. Therefore:

1. the nonperiodic logarithmic main term does not have an $o(1)$ error;
2. the quotient $F(x)/e^{M_0(x)}$ does not tend to 1;
3. no claimed error such as $O((\log x)^{-1})$ can hold after simply deleting Ψ .

The first harmonic is only about one part in a million, but a fixed one-part- in-a-million oscillation is still much larger than every error tending to zero.

17 The first saddle correction

The $O(1/\lambda)$ in Theorem 15.1 has an explicit periodic coefficient. Let primes denote ordinary derivatives of the one-periodic function Ψ . Define

$$A(u) = \frac{\Psi'(u)}{L} - \frac{1}{2}, \quad B(u) = -\frac{1}{4} + \frac{1}{2L} + \frac{\Psi'(u)}{2L} - \frac{\Psi''(u)}{2L^2}. \quad (17.1)$$

On the central scale $y = v/\sqrt{\lambda}$, the first exponent polynomials are

$$P_1(v; u) = i \left(A(u)v + \frac{v^3}{3} \right), \quad P_2(v; u) = \frac{v^4}{4} + B(u)v^2. \quad (17.2)$$

They come directly from expanding (15.6); the cubic term is a characteristic feature of the vertical Bromwich parametrization $s = r(1 + iy)$.

Definition 17.1. The first logarithmic saddle coefficient is

$$c_1(u) = \frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(u) + \Psi'(u)^2}{2L^2}. \quad (17.3)$$

It is continuous, bounded, smooth, and one-periodic.

Theorem 17.2 (First correction). As $x \downarrow 0$,

$$\log F(x) = M(x) + \frac{c_1(\lambda(x))}{\lambda(x)} + O\left(\frac{1}{\lambda(x)^2}\right). \quad (17.4)$$

Equivalently,

$$F(x) = e^{M(x)} \left[1 + \frac{c_1(\lambda(x))}{\lambda(x)} + O\left(\frac{1}{\lambda(x)^2}\right) \right]. \quad (17.5)$$

Proof. After factoring out the height (15.7), setting $y = v/\sqrt{\lambda}$, and retaining terms through λ^{-1} , the central integrand has the form

$$e^{-v^2/2} \left[1 + \frac{P_1(v; u)}{\sqrt{\lambda}} + \frac{P_2(v; u) + \frac{1}{2}P_1(v; u)^2}{\lambda} + O\left(\frac{(1+|v|)^9}{\lambda^{3/2}}\right) \right], \quad (17.6)$$

where $u = \lambda$ modulo 1. The odd term integrates to zero. If V is a standard normal random variable, the relative coefficient of $1/\lambda$ is

$$\mathbb{E} \left[P_2(V; u) + \frac{1}{2}P_1(V; u)^2 \right]. \quad (17.7)$$

Using $\mathbb{E}V^2 = 1$, $\mathbb{E}V^4 = 3$, and $\mathbb{E}V^6 = 15$, followed by the substitutions (17.1), reduces (17.7) to

$$\frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(u) + \Psi'(u)^2}{2L^2} = c_1(u).$$

The minor arcs and the forward tail are smaller than λ^{-2} relative to the main mass after a one-order refinement of the estimates in Theorem 15.1. This proves (17.5); taking the logarithm leaves the same first coefficient and proves (17.4). $\square$

Combining (16.8) and (17.4) gives the particularly useful formula

$$\begin{aligned} \log F(2^{-t}) = & -\frac{L}{2}t^2 - t \log t + \left(1 + \frac{L}{2}\right)t - \frac{(\log t)^2}{2L} + C_* + \Psi(\lambda(t)) \\ & - \frac{(\log t)^2}{2L^2t} + \frac{c_1(\lambda(t))}{\lambda(t)} + O\left(\frac{(\log t)^3}{t^2}\right). \end{aligned} \quad (17.8)$$

The periodic functions remain at the exact phase; only the elementary part has been expanded in t .

18 The full all-orders expansion

The saddle method does not stop at the first correction. The periodic Fourier series is differentiable to all orders, the forward tail is exponentially flat, and the central kernel admits a complete Gaussian expansion.

Theorem 18.1 (Full exact-phase expansion). *There exist bounded smooth one-periodic functions $c_j : \mathbb{R} \rightarrow \mathbb{R}$ for $j \geq 0$, with*

$$c_0 = 0, \quad c_1 \text{ given by (17.3)}, \quad (18.1)$$

such that, for every integer $N \geq 1$,

$$\log F(x) = M(x) + \sum_{j=1}^{N-1} \frac{c_j(\lambda(x))}{\lambda(x)^j} + O\left(\frac{1}{\lambda(x)^N}\right) \quad (x \downarrow 0). \quad (18.2)$$

Equivalently, the error is $O((- \log x)^{-N})$. The coefficient functions are evaluated at the exact lower-Lambert phase.

Proof. We outline the all-orders construction.

Central analytic expansion. In (15.6), put $y = v/\sqrt{\lambda}$ and expand $\delta(y) = \text{Log}(1 + iy)$, the analytic periodic function $\Psi(\lambda + \delta/L)$, and the amplitude $(1 + iy)^{-1}$ to an arbitrary fixed order. After the Gaussian term $-v^2/2$ is removed, one obtains exponent polynomials

$$\sum_{m=1}^{2N} \lambda^{-m/2} P_m(v; \lambda), \quad (18.3)$$

where every coefficient of P_m is a polynomial in finitely many derivatives of Ψ evaluated at λ . These coefficients are bounded and one-periodic.

Exponentiation and Gaussian contraction. Expanding the exponential of (18.3) produces finite Bell-polynomial combinations. Integrating termwise against $e^{-v^2/2}$ kills every odd total power. Thus only integral powers λ^{-j} survive. The resulting relative-mass coefficients $b_j(\lambda)$ are bounded smooth periodic functions, with $b_0 = 1$.

Uniform remainder. Choose a central window that grows as a small power of λ in the v -variable. Taylor's theorem and the exponential decay of the Gamma-zeta Fourier coefficients give an integrable Gaussian majorant for the truncated remainder. The intermediate and minor arcs satisfy estimates stronger than any prescribed inverse power after the truncation parameters are chosen in the usual order. The tail $E(2^\lambda)$ is exponentially small and may be added to an expansion of every finite order.

Taking the logarithm. The saddle ratio has an expansion $1 + \sum_{j=1}^{N-1} b_j(\lambda) \lambda^{-j} + O(\lambda^{-N})$. The formal power series for $\log(1 + z)$ converts the b_j into periodic polynomials c_j without changing the remainder order. The first one is $c_1 = b_1$, already computed in Theorem 17.2. This proves (18.2). $\square$

18.1 All-order expansion of the phase

The exact phase can itself be expanded in powers of $1/t$. Let $u = \log t$. Solving (14.5) recursively gives

$$\begin{aligned} \lambda(t) = t + \frac{u}{L} + \frac{u}{L^2 t} + \frac{u - u^2/2}{L^3 t^2} \\ + \frac{u - 3u^2/2 + u^3/3}{L^4 t^3} + O\left(\frac{(1+u)^4}{t^4}\right). \end{aligned} \quad (18.4)$$

More generally, there are unique polynomials $P_j(u)$ such that, for every N ,

$$\lambda(t) = t + \frac{u}{L} + \sum_{j=1}^{N-1} \frac{P_j(u)}{t^j} + O\left(\frac{(1+u)^N}{t^N}\right). \quad (18.5)$$

They are obtained by substituting the right side into $\lambda = t + L^{-1} \log \lambda$ and cancelling successive powers of $1/t$.

Remark 18.2 (Exact phase versus expanded phase). Theorem 18.1 deliberately keeps every periodic coefficient at $\lambda(x)$, not at a truncated approximation to it. One may combine (18.5) with Taylor expansions of Ψ and the c_j to any fixed order. A single all-orders statement after composing all periodic functions with a formal phase expansion requires a separate composition theorem and additional bookkeeping; it is not silently assumed here.

18.2 A numerical check at exact inverse powers

The values $F(2^{-n})$ are exact rationals, so they provide unusually clean tests of the asymptotic formula. The following table compares the exact logarithm with M ; the last column is the predicted first coefficient at the corresponding exact phase.

n	$\lambda(2^{-n})$	$\log F(2^{-n}) - M$	$\lambda(\log F - M)$	$c_1(\lambda)$
10	13.78503056	0.0580133344	0.79971559	0.76296785
20	24.62186833	0.0318275697	0.78365423	0.76292687
40	45.50804986	0.0170138895	0.77426893	0.76294905

The scaled residual moves toward the periodic first correction; the remaining difference is of order $1/\lambda$, as predicted by (17.4).

19 Consequences and conceptual summary

Several themes now fit together.

Self-similarity versus exact arithmetic. The global equation $\Theta'(x) = 2\Theta(2x)$ turns differentiation into a binary magnification. At dyadic points, sufficient magnification lands at an integer, where Θ is either zero or a Thue–Morse sign. This is why Taylor data terminate and why dyadic values are rational.

The moments are the bridge. The same c_n appear as Fourier coefficients near the origin, complete even moments of the bump, coefficients in the exact dyadic formula, and building blocks of the Legendre expansion. Their integer normalization $\mathcal{F}_n$ controls denominators and parity.

The endpoint remembers the discrete scale. The quadratic term in $\log F(x)$ comes from the triple pole at zero in the Mellin representation of the dyadic harmonic sum. The other poles on the imaginary axis encode the one-periodic fluctuation. The fluctuation is tiny because Gamma decays exponentially on vertical lines, but it is nonzero because neither Gamma nor zeta vanishes at the relevant points.

Lambert W is not cosmetic. The equation $\lambda 2^{-\lambda} = x$ exactly balances the Laplace radius and the endpoint argument. Replacing λ too early by $\log_2(1/x)$ loses terms involving $\log \log(1/x)$ and misplaces the phase of Ψ . The exact lower branch is therefore the natural coordinate for the full expansion.

Smoothness is extreme but intermediate. The Fabius function is flatter than every power at the endpoint but much larger than $\exp(-x^{-\alpha})$ for every $\alpha > 0$. Its Taylor series is zero at the endpoint and finite at every reduced dyadic point, yet it is nowhere analytic. The asymptotic formula supplies a quantitative profile for this otherwise qualitative pathology.

A Infinite binary-reduction series and discrete limits

The finite dyadic formulas of the main text terminate because the binary tail of a dyadic number eventually vanishes. For a general real argument the same Taylor reductions do not terminate, but they telescope. This appendix states the resulting absolutely convergent series, including the scale-zero term that is easy to overlook, and then records the related discrete-limit formula.

A.1 Binary prefixes, tails, and the analytic telescope

For $x \geq 0$ and $m \in \mathbb{N}$, put

$$A_m = A_m(x) = \lfloor 2^m x \rfloor, \quad Y_m = Y_m(x) = x - \frac{A_m}{2^m}, \quad (\text{A.1})$$

and define the previous prefix by

$$B_0 = B_0(x) = \left\lfloor \frac{x}{2} \right\rfloor, \quad B_m = B_m(x) = A_{m-1}(x) \quad (m \geq 1). \quad (\text{A.2})$$

Thus $0 \leq Y_m < 2^{-m}$. The integer $A_m - 2B_m \in \{0, 1\}$ is the m th newly exposed binary digit, with the scale-zero convention built into (A.2).

Define the finite reduction polynomial

$$\mathcal{R}_m(y) = \sum_{k=0}^m 2^{\binom{k+1}{2} - \binom{m-k}{2}} \frac{d_{m-k}}{(m-k)!} \frac{y^k}{k!}. \quad (\text{A.3})$$

This is exactly the Taylor polynomial appearing in (8.2), now indexed by the scale m .

Theorem A.1 (Global binary-reduction series). *For every $x \geq 0$,*

$$\Theta(x) = \sum_{m=0}^{\infty} \epsilon_{A_m} (2B_m - A_m) \mathcal{R}_m(Y_m), \quad (\text{A.4})$$

and the series is absolutely convergent. If $0 \leq x \leq 1$, the left side may be replaced by the bounded Fabius function $F(x)$.

Proof. Let

$$E_m(x) = \epsilon_{A_m} \Theta(Y_m). \quad (\text{A.5})$$

The one-step Taylor reduction says that passing from the $(m-1)$ st tail to the m th tail either changes nothing, when the new binary digit is 0, or crosses one inverse-power center and contributes the polynomial $\mathcal{R}_m$. In both cases the identity can be written uniformly as

$$E_{m-1}(x) - E_m(x) = \epsilon_{A_m} (2B_m - A_m) \mathcal{R}_m(Y_m) \quad (m \geq 1). \quad (\text{A.6})$$

At scale zero, the same decomposition is valid after first locating x in its even or odd block of length one:

$$\Theta(x) = \epsilon_{A_0} (2B_0 - A_0) \mathcal{R}_0(Y_0) + E_0(x). \quad (\text{A.7})$$

Summing (A.6) and using (A.7) gives the exact finite telescope

$$\Theta(x) = \sum_{m=0}^N \epsilon_{A_m} (2B_m - A_m) \mathcal{R}_m(Y_m) + E_N(x). \quad (\text{A.8})$$

Since $0 \leq Y_N < 2^{-N}$ and $F(y) \leq 2y$ on $[0, 1/2]$, for $N \geq 1$ we have

$$|E_N(x)| \leq 2^{1-N}. \quad (\text{A.9})$$

Hence $E_N(x) \rightarrow 0$, proving ordinary convergence. Moreover, (A.6) and the same bound give, for $m \geq 2$,

$$|\epsilon_{A_m} (2B_m - A_m) \mathcal{R}_m(Y_m)| \leq 4 \cdot 2^{-(m-1)}. \quad (\text{A.10})$$

The two initial terms are finite and the geometric majorant proves absolute convergence. On the closed unit interval, $\Theta = F$. $\square$

A.2 The parity-power form and the missing endpoint term

The inverse-power bridge (6.13) rewrites half of the reduction polynomial as

$$\mathcal{I}_m(y) = \sum_{n=0}^m 2^{\binom{n+1}{2}-1} \Theta(2^{n-m}) \frac{y^n}{n!} = \frac{1}{2} \mathcal{R}_m(y). \quad (\text{A.11})$$

The exponent for $n = 0$ is genuinely -1 . With the usual convention $0^0 = 1$, no exceptional interpretation is required.

Corollary A.2 (Parity-power series). *For $x \geq 0$,*

$$\Theta(x) = \sum_{m=0}^{\infty} ((-1)^{A_m} - 1) \epsilon_{A_m} \mathcal{I}_m(Y_m), \quad (\text{A.12})$$

and the series is absolutely convergent.

Proof. For every nonnegative integer A ,

$$\frac{(-1)^A - 1}{2} = 2 \left\lfloor \frac{A}{2} \right\rfloor - A. \quad (\text{A.13})$$

Since $B_m = \lfloor A_m/2 \rfloor$, including $m = 0$, multiplication of (A.11) by (A.13) turns the summand in (A.12) into the summand in (A.4). $\square$

Remark A.3 (Why the index must begin at zero). For $0 \leq x < 1$, $A_0 = 0$, so the $m = 0$ term in (A.12) vanishes. This explains why a formerly one-indexed version happens to work on the half-open interval. At $x = 1$, however,

$$A_0 = 1, \quad Y_0 = 0, \quad \mathcal{I}_0(0) = 2^{-1} \Theta(1) = \frac{1}{2},$$

and the omitted term equals $((-1)^1 - 1) \epsilon_1 \mathcal{I}_0(0) = 1$. Thus the scale-zero term is indispensable at the right endpoint and, globally, distinguishes the even and odd unit blocks of Θ .

A.3 The fully explicit global q -binomial series

For a scalar $q \in \mathbb{R}$ or $q \in \mathbb{C}$, define

$$\mathcal{C}_n(q) = \frac{1}{2^{\binom{n}{2}}(1/2; 1/2)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} \epsilon_r (r - 2^k + q)^{n+k}, \quad (\text{A.14})$$

and

$$\mathcal{P}_m(q; z) = \frac{1}{2^{\binom{m+1}{2}}} \sum_{n=0}^m \frac{z^{m-n}}{(m-n)!} \mathcal{C}_n(q). \quad (\text{A.15})$$

Although the right side is written using q , it is a constant polynomial in that common translation.

Theorem A.4 (Global q -binomial–Thue–Morse series). *Let $x \geq 0$, and let q be any real or complex number. Then*

$$\Theta(x) = \sum_{m=0}^{\infty} \epsilon_{A_m} (2B_m - A_m) \mathcal{P}_m(q; 2^{m+1}Y_m). \quad (\text{A.16})$$

For complex q the equality is understood after embedding the real number $\Theta(x)$ into $\mathbb{C}$. The series is absolutely convergent, and every single summand is independent of q before summation. On $[0, 1]$ the sum equals $F(x)$.

Proof. The finite translation-invariance argument of [Section 9](#) applies to the inner polynomial at each fixed scale and gives the pointwise identity

$$\mathcal{P}_m(q; 2^{m+1}y) = \mathcal{R}_m(y). \quad (\text{A.17})$$

One way to see the normalization is to expand the left side by (A.14)–(A.15): the Gaussian-binomial annihilator removes every translation term of degree below the active order, and the surviving coefficient is precisely the inverse-power Taylor datum in (A.3). Substitution of (A.17) into (A.4) proves the identity and imports absolute convergence. Since the equality is polynomial, the rational proof extends without change to real and complex translations. $\square$

A.4 A finite discrete approximation converging to Θ

For $x \geq 0$, let

$$N_p(x) = \left\lfloor 2^p x + \frac{1}{2} \right\rfloor. \quad (\text{A.18})$$

Equivalently, the inner sum below has inclusive upper endpoint $\lfloor 2^p x - 1/2 \rfloor$; the formulation using N_p also covers the empty case safely. For $q \in \mathbb{R}$ or $q \in \mathbb{C}$, define

$$\begin{aligned} D_n(q, x) &= \frac{1}{2^{n^2}(1/2; 1/2)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \\ &\quad \times \sum_{r=0}^{N_{n+k}(x)-1} \epsilon_r (r - 2^{n+k}x + q)^{n+k}. \end{aligned} \quad (\text{A.19})$$

An empty inner range contributes zero.

Theorem A.5 (Discrete-limit formula). *For every fixed $x \geq 0$ and every fixed real or complex q ,*

$$\lim_{n \rightarrow \infty} D_n(q, x) = \Theta(x). \quad (\text{A.20})$$

In particular, the limit is $F(x)$ for $0 \leq x \leq 1$ and is independent of the translation q .

Proof. The proof separates an exact algebraic reindexing from two convergence estimates.

Put $j = n - k$. After normalizing the finite Thue–Morse power sum as a spline branch of degree $2n - j$, the outer coefficients become the triangular array

$$w_{n,j} = \frac{(-1)^j [n]_{1/2}^{[j]} 2^{-\binom{j+1}{2}}}{(1/2; 1/2)_n}, \quad 0 \leq j \leq n. \quad (\text{A.21})$$

The finite q -binomial theorem gives the exact row-sum identity

$$\sum_{j=0}^n w_{n,j} = 1. \quad (\text{A.22})$$

The same theorem with the absolute signs removed only after taking norms shows the uniform variation bound

$$\sum_{j=0}^n |w_{n,j}| \leq 16. \quad (\text{A.23})$$

Every branch index sampled in the n th row tends uniformly to infinity, and the real, unshifted spline branches converge to $\Theta(x)$. Therefore the finite-row Toeplitz lemma, using (A.22)–(A.23), shows that their weighted average tends to $\Theta(x)$.

It remains to restore q . On a fixed real branch, translation has the exact finite Taylor expansion

$$H_{p,M}(z + q) - H_{p,M}(z) = \sum_{d=0}^{p-1} \frac{2^{\binom{d}{2}}}{2^{\binom{p}{2}} (p-d)!} q^{p-d} H_{d,M}(z), \quad (\text{A.24})$$

where $H_{p,M}$ denotes the normalized degree- p Thue–Morse branch with prefix length M . The lower real branches have norm at most one in the relevant range, whence

$$|H_{p,M}(z + q) - H_{p,M}(z)| \leq 2^{-(p-1)} e^{|q|}. \quad (\text{A.25})$$

Since $p = 2n - j \geq n$ in every row, this perturbation tends to zero uniformly in j . Combining it with the bounded total variation proves (A.20) over both $\mathbb{R}$ and $\mathbb{C}$. $\square$

Remark A.6. The finite approximants $D_n(q, x)$ can depend on q ; only their limit is translation-independent. This differs from Theorem A.4, where each fixed-scale polynomial is already constant in q .

B Concordance with the formal development and source corrections

The article is organized mathematically rather than by Lean file. The following concordance indicates where the principal source groups have been translated into ordinary analysis.

Formal or printed source group	Human-readable treatment in this article
Arias de Reyna, first paper: existence, uniqueness, convolution, and probability model	Section 2 and the construction in the sections beginning with “Construction by probability, convolution, and Fourier transform.”
First paper: polynomial step approximants and their pointwise limit	Section 4 , including the half-weight endpoint convention.
First paper: partition of unity, Poisson summation, cosine product, and Weierstrass product	The latter half of Section 4 .
First paper: derivative identities, integer values, non-analyticity, and moments	Section 5 and the moment section beginning on page 11.
First paper: rationality and the finite dyadic formula	Theorem 7.2 and its proof in the section “A closed formula for every dyadic value.”
Arithmetic paper: c_n, d_n , their generating functions and recurrences	The moment section, especially (6.3), (6.7), and (6.10).
Arithmetic paper: exact inverse powers, Taylor recursion, and the executable evaluator	Section 8 .
Arithmetic paper: integer normalizations, denominator bounds, parity, valuations, and Bernoulli recurrences	Section 10 ; the still open denominator equality is explicitly kept as a conjecture.
Additional formal development: finite translated q -binomial formula	Section 9 .
Additional formal development: Fourier–Legendre coefficients and uniform series	Section 11 .
Additional formal development: negative Laplace product, Mellin finite part, Gamma–zeta fluctuation, exact Lambert phase, saddle method, and all orders	The endpoint sequence of sections from “The negative Laplace product” through Section 18 .
Additional formal development: binary telescope, corrected infinite q -binomial series, parity-power series, and discrete limit	Section A .

B.1 Corrections incorporated in the statements

Several printed or informal formulas require small but mathematically important repairs. They have been incorporated silently into the main statements and are listed here for auditability.

1. The convolution occurring in the first paper's equation (12) is finite; an infinite upper

index is incompatible with the displayed dependence on the stage parameter.

2. Closed interval indicators in the early step approximants double-count shared endpoints. The correct normalized approximants give each shared endpoint weight $1/2$.
3. In the Poisson formulas, the exponential phase retains the translation parameter, the differentiated identity requires positive order, and the closed dyadic formula uses the consistent scaling adopted in [Theorem 7.2](#).
4. The bounded CDF-style function satisfies $F'(x) = 2F(2x)$ only on its left half. The equation valid on all of $\mathbb{R}$ belongs to the signed extension Θ .
5. The translated derivative lemma in the arithmetic paper requires the nonnegative scale-plus-order hypothesis used in its proof.
6. In the integer R_n , the power of two is the positive exponent consistent with the recurrence, proof, and displayed numerical values.
7. The naturality identity used in the parity argument contains the factor of two forced by the normalization of the half moments.
8. The global binary and q -binomial series begin at $m = 0$, with $B_0 = \lfloor x/2 \rfloor$. The omitted term vanishes on $[0, 1)$ but equals one at $x = 1$.
9. A sharp endpoint formula cannot discard the Gamma–zeta periodic fluctuation. Its nonzero Fourier modes are present at the exact lower-Lambert phase; replacing that phase prematurely also displaces terms involving $\log \log(1/x)$.

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